PORTFOLIO PROJECT · SAS · UNSUPERVISED LEARNING

Finding patient risk groups

PCA and cluster analysis on heart disease patient data, built in SAS

Dhruv Khanna · MS Business Analytics, Quinnipiac University

303 patients · 12 variables

THE DATA

303 patients, 12 clinical measures, nothing missing

54.4
mean age, range 29 to 77

131.7
mean resting blood pressure, mm Hg

246.7
mean cholesterol, mg/dL

149.6
mean maximum heart rate, bpm

1.04
mean ST depression on exercise

Age rises with ST depression and falls with maximum heart rate. Blood pressure and cholesterol barely correlate with anything else.

THE APPROACH

Four steps from raw measures to patient groups

Step 1

Profile

Distributions, frequencies and correlations

Step 2

Reduce

PCA to find the main dimensions of variation

Step 3

Segment

Hierarchical and k-means clustering

Step 4

Describe

Profile each cluster and compare methods

PRINCIPAL COMPONENT ANALYSIS

Heart risk here has no single dominant dimension

34.3% of the variance is explained by the first two components. Maximum heart rate and ST depression contribute the most to them.

CLUSTER ANALYSIS

Two patient groups, found by both methods

Cluster 1 · Older, higher risk

- Mean age 58 to 59
- Resting blood pressure 134 to 138 mm Hg
- ST depression 1.7
- More men, more severe chest pain

Cluster 2 · Younger, lower risk

- Mean age 51
- Maximum heart rate 161 to 163 bpm
- ST depression 0.37
- Mostly women, milder chest pain

Ranges span the hierarchical and k-means results.

SO WHAT

What a care provider could do, and what I'd test next

Act

- Prioritize Cluster 1 for check-ups and coaching
- Tailor outreach by age and risk group
- Flag new patients who match Cluster 1

Test next

- Score cluster quality with silhouette values
- Check whether clusters match actual diagnoses
- Then build a supervised model to predict risk